

CS560 – Software Engineering
Project Proposal & SRS: Kraftify



Submitted By:

Muhammad Sannan - 002356065

Reza Nobahari – 002353688

Hassan Rhezouane - 002345731

Proposal: Digital Platform for Skilled Tradespeople

1. Problem Statement

Millions of skilled tradespeople plumbers, electricians, carpenters, and home service providers remain invisible in Canada's digital economy.

Despite having real talent and strong local demand, most of them still operate offline or under someone else's business name, leaving them without:

Credibility there's no verified digital identity or reliable rating system that showcases the quality of their work.

Direct access to customers mostly rely on word of mouth or middlemen who take a large portion of their earnings.

Because of this, talented workers struggle to find consistent jobs, while customers face the challenge of finding trustworthy and verified professionals.

2. Proposed Solution

We are creating a digital marketplace designed to help skilled workers build credibility and connect directly with customers.

Each tradesperson will have a verified digital profile where they can highlight their skills, certifications, past projects, and customer reviews.

Customers can easily discover, hire, and pay trusted professionals through a simple and secure interface.

This platform removes the need for intermediaries, ensuring fair earnings for workers and creating direct, transparent relationships between tradespeople and their customers.

3. Scope

Kraftify is a **web application** supporting three roles: **Customer**, **Tradesperson (Worker)**, and **Administrator**. Core capabilities include verified worker profiles, job posting and discovery, booking, messaging, reviews, and admin verification/dispute handling.

4. Target Users

Primary Users

Skilled Tradespeople: plumbers, electricians, carpenters and cleaners who want verified digital identities and independence.

Secondary Users

Homeowners and Small Businesses: people looking for reliable, transparent, and easy to book local professionals.

5. Functional Requirements

5.1.1 Customer Requirements

- **FR1:** Register, log in, and manage profile.
- **FR2:** Post a job (description, location, desired time).
- **FR3:** Search/filter workers by trade, location radius, and rating.
- **FR4:** Book a worker and communicate via in-app chat.
- **FR5:** Submit a rating/review after job completion.

5.1.2 Tradesperson (Worker) Requirements

- **FR6:** Create/edit professional profile (skills, rates, portfolio).
- **FR7:** Upload certifications for admin verification.
- **FR8:** Receive/accept/decline job requests; track job status and earnings.

5.1.3 Administrator Requirements

- **FR9:** Verify worker identities/certifications; set verified status.
- **FR10:** Moderate disputes and user reports.
- **FR11:** Access audit logs for key actions.

5.2 Non-Functional Requirements

Performance & UX

- **NFR1:** Fast and responsive interactions for search, booking, and messaging.

Security

- **NFR2:** HTTPS/TLS for all traffic; **JWT** for authentication
- **NFR3:** Passwords hashed; sensitive fields encrypted at rest.

Reliability & Availability

- **NFR4:** Target $\geq 99\%$ availability and regular automated backups

Logging and Monitoring

- **NFR5:** The system should maintain detailed logs for all critical actions (login, booking, posting, etc.).

6. Technology Stack

Cloud-hosted client–server system exposing RESTful JSON APIs.

Frontend: **React**. Backend: **Python (Flask/FastAPI)**. Database: **PostgreSQL**.

7. Workflow

Customer Posts a Job

1. Customer posts a job with details.
2. Worker applies or is matched based on location and skill.
3. Customer reviews profile and confirms booking.
4. Both communicate via chat.
5. Worker completes job → customer reviews → worker rating updates.

Customer Searches for a Skilled Worker

1. The customer searches for a skilled worker using filters such as skill type, location, rating, or availability.
2. The platform displays verified worker profiles with portfolios, certifications, and reviews.
3. The customer views details, compares profiles, and initiates direct contact or booking through the platform.
4. Both parties discuss the project, agree on terms, and proceed with the service.
5. Upon completion, the customer leaves a review which updates the worker's overall rating.